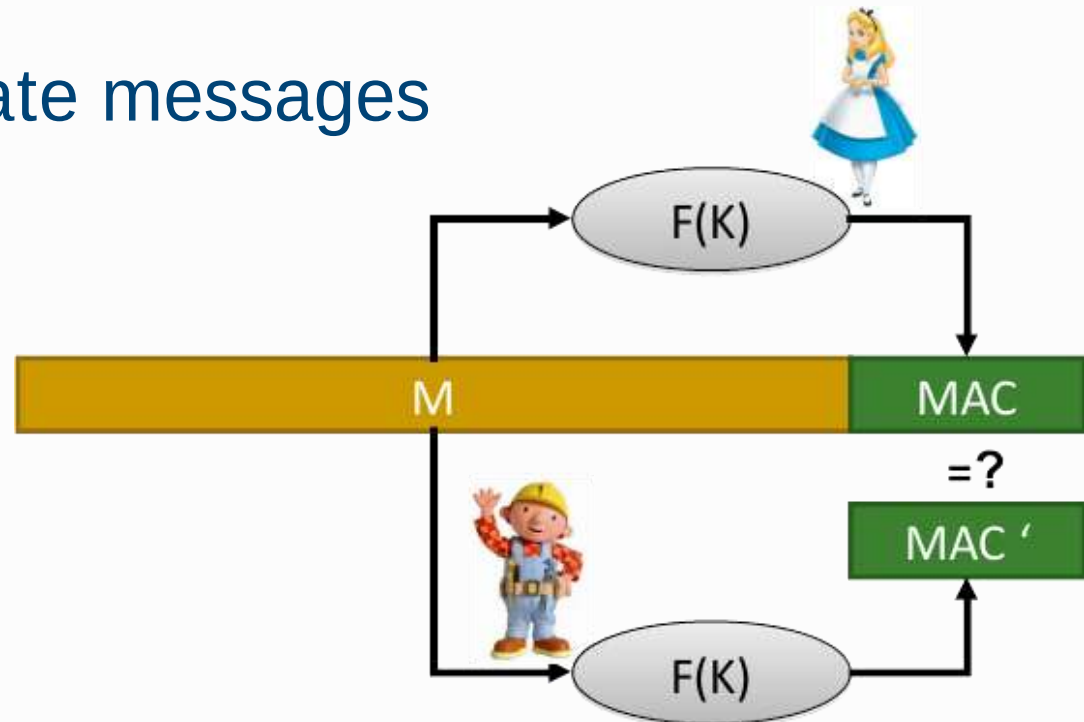


Message Authentication Codes

Message Authentication Code (MAC)

- ▶ MIC computed with a key
 - ◆ Only key holders can generate/validate the MAC
- ▶ Used to authenticate messages
 - ◆ $M' = M \mid \text{MAC}(M)$



MAC: approaches

- ▷ Encryption of an ordinary digest
 - ◆ Using, for instance, a symmetric block cipher
- ▷ Using encryption with feedback & error propagation
 - ◆ CBC-MAC
- ▷ Adding a key to the hashed data
 - ◆ HMAC (output length depends on the function H used)
 - $H(K, \text{opad}, H(K, \text{ipad}, \text{text}))$
 - $\text{ipad} = 0x36 \text{ B times}$ $\text{opad} = 0x5C \text{ B times}$ $B = \text{size of H input block}$
 - HMAC-MD5, HMAC-SHA-1, etc.

Encryption + Authentication

- ▷ Encrypt-then-MAC: MAC is computed from cryptogram
 - ◆ Allows verifying integrity before (the longer) decryption
 - ◆ Preferable option
- ▷ Encrypt-and-MAC: MAC is computed from plaintext
 - ◆ MAC is not encrypted
 - ◆ May give information regarding original text (if similar to other)
- ▷ MAC-then-Encrypt: MAC is computed from plaintext
 - ◆ MAC is encrypted
 - ◆ Requires full decryption before MAC is validated

GCM (Galois Counter Mode)

